```
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#BSAN 360
#Project Assignments

#Part 1
#1
import pandas as pd
amazon = pd.read_csv("amazon_sales_dataset.csv")
amazon
```

	order_id	order_date	product_id	product_category	price	discount_percent	quantity_sold	customer_region	payı
0	1	2022-04-13	2637	Books	128.75	10	4	North America	
1	2	2023-03-12	2300	Fashion	302.60	20	5	Asia	
2	3	2022-09-28	3670	Sports	495.80	20	2	Europe	
3	4	2022-04-17	2522	Books	371.95	15	4	Middle East	
4	5	2022-03-13	1717	Beauty	201.68	0	4	Middle East	
...	...	...	...	...	...	...	...	...	
49995	49996	2022-09-03	1433	Beauty	26.99	0	5	Middle East	
49996	49997	2022-07-03	1428	Beauty	294.23	10	5	Asia	
49997	49998	2023-02-17	4651	Electronics	352.11	30	4	Asia	
49998	49999	2022-09-30	4371	Beauty	307.54	5	1	Middle East	
49999	50000	2023-06-29	2944	Home & Kitchen	253.44	30	1	Europe	

50000 rows × 13 columns

Next steps: Generate code with amazon New interactive sheet

```
#2
amazon.shape
#There is 50000 rows and 13 columns in this dataset; Everything looks correct
```

```
(50000, 13)
```

```
#3
amazon.head()
amazon.tail()
#Everything looks correct
```

	order_id	order_date	product_id	product_category	price	discount_percent	quantity_sold	customer_region	payı
49995	49996	2022-09-03	1433	Beauty	26.99	0	5	Middle East	
49996	49997	2022-07-03	1428	Beauty	294.23	10	5	Asia	
49997	49998	2023-02-17	4651	Electronics	352.11	30	4	Asia	
49998	49999	2022-09-30	4371	Beauty	307.54	5	1	Middle East	
49999	50000	2023-06-29	2944	Home & Kitchen	253.44	30	1	Europe	

```
#4
amazon.columns
#Everything was inputted correctly
```

```
Index(['order_id', 'order_date', 'product_id', 'product_category', 'price',
       'discount_percent', 'quantity_sold', 'customer_region',
       'payment_method', 'rating', 'review_count', 'discounted_price',
       'total_revenue'],
      dtype='object')
```

```
#5a
df_q1 = amazon[["order_date", "price", "customer_region", "total_revenue"]]
df_q1
#Some example of what the column provide is the order date they purchase the product,
```

```
#the price is the cost of the product, and customer region is the region they're from, and total revenue
#is amount generated from that product within that period of time.
#5b
#The data type for Order_Date is datetime which tells us when each order was placed and analyze sales and
#customer behavior overtime
#The data type for Price is the numeric (real number) which is the price of the product
#The data type for Customer_Region is the character string, which helps us associate where the customer is
#from and helps us analyze/compare the sales across the regions.
#The data type for total_revenue is numeric. The revenue is amount generated from the order, typically
#calculated by the price*quantity
#5c
amazon["order_date"].min()
amazon["order_date"].max()
amazon["price"].min()
amazon["price"].max()
amazon["customer_region"].unique()
amazon["customer_region"].mode()
amazon["total_revenue"].min()
amazon["total_revenue"].max()
#5d
amazon["order_date"].isna().sum()
#There isn't any missing values in any of the columns
```

```
np.int64(0)
```

```
#6
#In this dataset, I will analyze the Amazon Sales Dataset to identify trends and patterns of sales and
#customers over the period. This includes categories such as the items they bought, best performing products,
#price of product, and how sales vary from season to season. Some research questions I have is
#Q1.Do average sales differ significantly across different regions?
#Some columns I need to answer this question is the order date, price, customer region, total revenue
#Q2.Are there statistically significant differences in average sales across product categories?
#Some columns I need to answer this question is the order date, price, total revenue, product category
#Q3.Is sales growth consistent across regions over time?
#Some columns I need to answer this question is the order date, price, customer region, total revenue, and quantity
#sold
```

```
#7
df_q1 = amazon[["order_date", "price", "customer_region", "total_revenue"]]
df_q1
df_q2 = amazon[["order_date", "price", "product_category", "total_revenue"]]
df_q2
df_q3 = amazon[["order_date", "price", "quantity_sold", "customer_region", "total_revenue"]]
df_q3
```

	order_date	price	quantity_sold	customer_region	total_revenue
0	2022-04-13	128.75	4	North America	463.52
1	2023-03-12	302.60	5	Asia	1210.40
2	2022-09-28	495.80	2	Europe	793.28
3	2022-04-17	371.95	4	Middle East	1264.64
4	2022-03-13	201.68	4	Middle East	806.72
...	...	...	...	...	...
49995	2022-09-03	26.99	5	Middle East	134.95
49996	2022-07-03	294.23	5	Asia	1324.05
49997	2023-02-17	352.11	4	Asia	985.92
49998	2022-09-30	307.54	1	Middle East	292.16
49999	2023-06-29	253.44	1	Europe	177.41

50000 rows × 5 columns

Next steps: Generate code with df_q3 New interactive sheet

```
#8
#For Questions 1 and 2, I believe ANOVA would be helpful to help and test whether region influences sales
#outcomes, and compares mean sales across multiple categories.
```

```
#For Question 3, I believe we can use a regression model using sales, time and region to test whether trends
#differ between regions.
```

```
#Project Assignment 2
amazon = amazon[["order_date", "price", "quantity_sold", "product_category", "customer_region", "total_revenue"]]
amazon = amazon.rename(columns={"order_date": "Order_Date"})
amazon = amazon.rename(columns={"product_category": "Product_Category"})
amazon = amazon.rename(columns={"price": "Price"})
amazon = amazon.rename(columns={"quantity_sold": "Quantity_Sold"})
amazon = amazon.rename(columns={"customer_region": "Customer_Region"})
amazon = amazon.rename(columns={"total_revenue": "Total_Revenue"})
amazon["Order_Date"] = pd.to_datetime(amazon["Order_Date"], errors="coerce")
amazon_2023_first6 = amazon[(amazon["Order_Date"] >= "2023-01-01") & (amazon["Order_Date"] <= "2023-07-01")]
amazon_2023_first6 = amazon_2023_first6.sort_values(by="Order_Date",ascending=True)
amazon_2023_first6
```

	Order_Date	Price	Quantity_Sold	Product_Category	Customer_Region	Total_Revenue
8844	2023-01-01	67.98	5	Electronics	North America	288.90
27119	2023-01-01	107.93	1	Home & Kitchen	Europe	86.34
20147	2023-01-01	462.19	2	Electronics	North America	647.06
5959	2023-01-01	77.68	5	Beauty	North America	349.55
7289	2023-01-01	31.21	4	Sports	Europe	99.88
...	...	...	...	...	...	...
25265	2023-07-01	245.29	3	Electronics	North America	625.50
25386	2023-07-01	75.97	1	Fashion	North America	64.57
35648	2023-07-01	239.95	3	Books	Asia	683.85
37097	2023-07-01	17.50	2	Home & Kitchen	Europe	28.00
10435	2023-07-01	172.43	3	Sports	North America	362.10

12511 rows × 6 columns

Next steps: Generate code with amazon_2023_first6 New interactive sheet

```
#Project Assignment 3
#No merging or combining of datasets was required because all relevant
#variables were contained within a single file.
#All columns are capitalized
#No missing values
#First 6 months are filtered
#Date column converted to datetime
amazon.dtypes
amazon.isnull().sum()
pivot_region = amazon.pivot_table(
    index="Order_Date",
    columns="Customer_Region",
    values="Total_Revenue",
    aggfunc="sum"
)
pivot_region

pivot_category = amazon.pivot_table(
    index="Product_Category",
    columns="Customer_Region",
    values="Total_Revenue",
    aggfunc="sum"
)
pivot_category

melt = pivot_region.reset_index().melt(
    id_vars="Order_Date",
    var_name="Customer_Region",
    value_name="Total_Revenue"
)
melt

grouped = amazon.groupby(["Order_Date", "Customer_Region"])["Total_Revenue"].sum()
grouped
```

```
wide = grouped.unstack()
wide
```

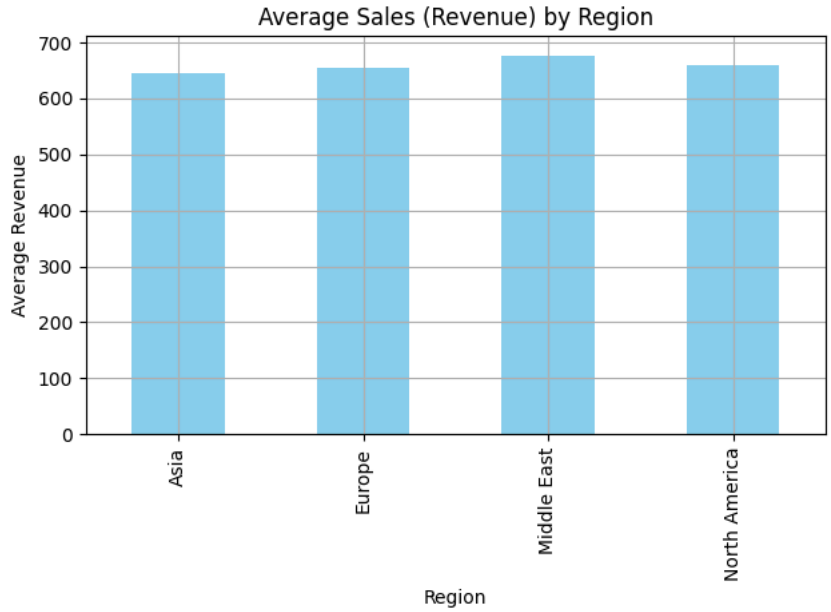
Customer_Region	Asia	Europe	Middle East	North America
Order_Date				
2022-01-01	12577.10	16190.37	7827.28	11265.62
2022-01-02	9729.73	9897.88	4334.81	11537.13
2022-01-03	9364.06	11433.25	10991.41	9141.15
2022-01-04	9406.61	9883.36	8760.42	6763.34
2022-01-05	11261.81	4751.22	15166.83	7136.86
...	...	...	...	...
2023-12-27	17612.57	10060.03	11930.14	8301.24
2023-12-28	17175.50	9780.17	8089.76	10194.20
2023-12-29	9689.87	9369.91	12546.96	12405.31
2023-12-30	10841.29	7135.88	11698.08	12288.25
2023-12-31	11816.48	13068.71	17583.11	13289.08

730 rows × 4 columns

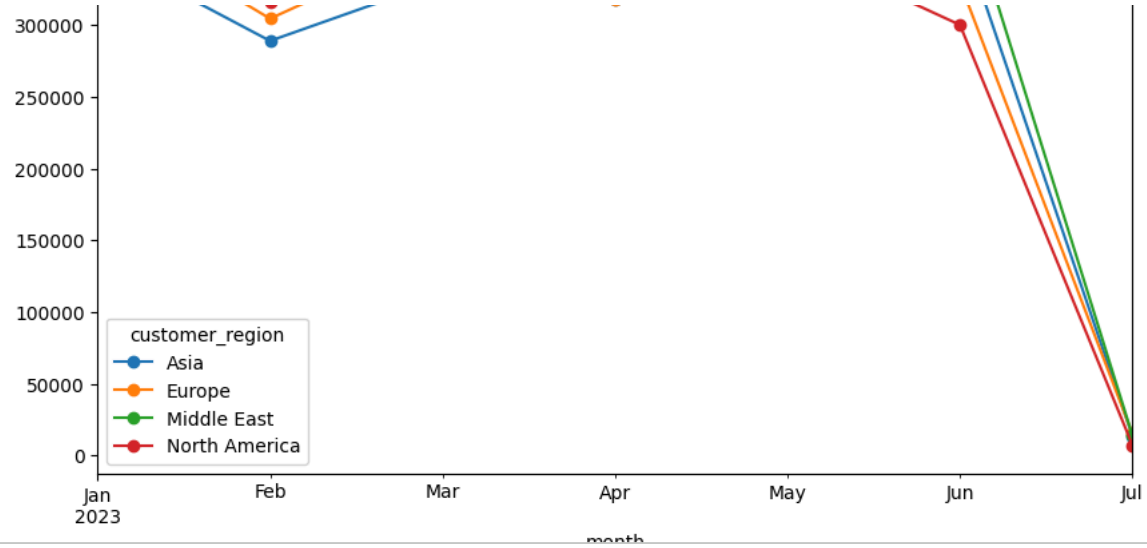
Next steps: Generate code with wide New interactive sheet

```
#Project Assignment 4

import pandas as pd
import matplotlib.pyplot as plt
amazon = pd.read_csv("amazon_sales_dataset.csv")

#Q1
region_means = amazon_2023_first6.groupby("Customer_Region")["Total_Revenue"].mean()
region_means.plot(kind='bar', color='skyblue')
plt.title("Average Sales (Revenue) by Region")
plt.xlabel("Region")
plt.ylabel("Average Revenue")
plt.tight_layout()
plt.grid(True)
plt.show()

#Average sales do not differ significantly across the different regions. The mean is relatively the same, but Middle
#the highest sales within the 6 months of 2023.

#Q2
category_means = amazon_2023_first6.groupby("Product_Category")["Total_Revenue"].mean()
category_means.plot(kind='bar')
category_means.plot(kind='bar', color='skyblue')
plt.title("Average Sales (Revenue) by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Average Revenue")
plt.tight_layout()
plt.grid(True)
plt.show()

#There isn't much difference in average sales across product cateogories. Electronics seems to have the highest sale
#category.

#Q3
amazon.columns = amazon.columns.str.strip().str.lower().str.replace(" ", "_")
amazon['order_date'] = pd.to_datetime(amazon['order_date'], errors='coerce')
amazon_2023_first6 = amazon[(amazon['order_date'] >= '2023-01-01') & (amazon['order_date'] <= '2023-07-01')].copy()
amazon['order_date'] = pd.to_datetime(amazon['order_date'])
amazon_2023_first6['month'] = amazon_2023_first6['order_date'].dt.to_period('M').dt.to_timestamp()

monthly_sales = amazon_2023_first6.groupby(['month', 'customer_region'])['total_revenue'].sum().unstack()
monthly_sales.plot(marker='o', figsize=(10,6))
monthly_sales.plot(kind='bar', color='skyblue')
plt.title("Sales Growth by Region Over Time (Jan–Jun 2023)")
plt.xlabel("Month")
plt.ylabel("Total Revenue")
plt.tight_layout()
```

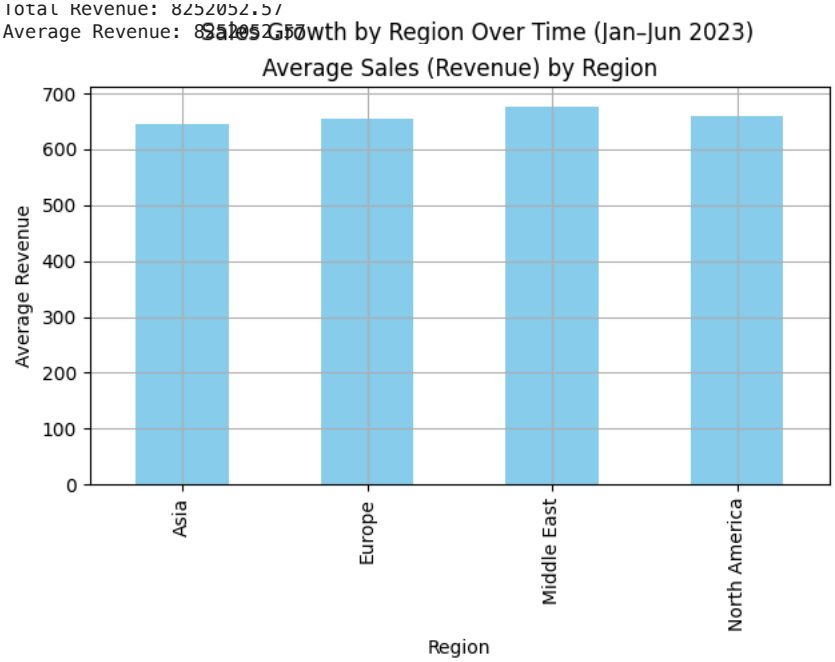
```
plt.grid(True)
plt.show()

#Sales growth is pretty consistent throughout the region based off the graph. In June, Middle East spent the most ou
#North America spent the least.
```

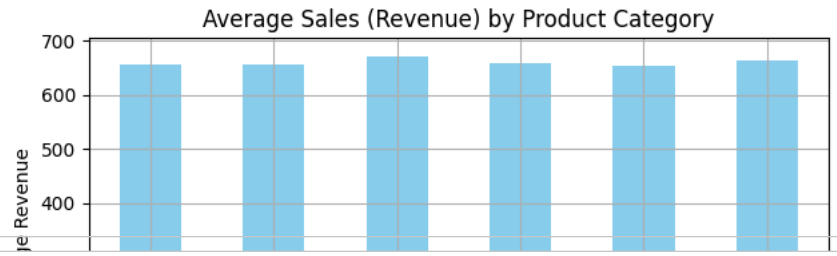



```
#Project Assignment 5
total_sales = amazon_2023_first6['total_revenue'].sum()
average_sales = amazon_2023_first6['total_revenue'].sum()
print("Total Revenue:", total_sales)
print("Average Revenue:", average_sales)

average_sales_means = amazon_2023_first6.groupby('customer_region')["total_revenue"].mean()
average_sales_means.plot(kind='bar', color='skyblue')
plt.title("Average Sales (Revenue) by Region")
plt.xlabel("Region")
plt.ylabel("Average Revenue")
plt.tight_layout()
plt.grid(True)
plt.show()

average_sales_category = amazon_2023_first6.groupby("product_category")["total_revenue"].mean()
average_sales_category.plot(kind='bar', color='skyblue')
plt.title("Average Sales (Revenue) by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Average Revenue")
plt.tight_layout()
plt.grid(True)
plt.show()
average_sales_category
average_sales_means

cross_tab = pd.crosstab(amazon_2023_first6['customer_region'],amazon_2023_first6['product_category'])
print(cross_tab)
#This shows how many orders occur in each region for each category
```



```
Total Revenue: 8252052.57
Average Revenue: 8252052.57
```






```
#Project Assignment 6

#Q1
import pandas as pd
amazon['order_date'] = pd.to_datetime(amazon['order_date'])
amazon_2023_first6 = amazon[(amazon["order_date"] >= "2023-01-01") & (amazon["order_date"] <= "2023-07-01")].copy()
amazon_2023_first6['month'] = amazon_2023_first6['order_date'].dt.to_period('M')

import statsmodels.api as sm
import statsmodels.formula.api as smf
model_region = smf.ols('total_revenue ~ C(customer_region)',data=amazon_2023_first6).fit()
anova_region = sm.stats.anova_lm(model_region, typ=2)
print(anova_region)

#Q1.Do average sales differ significantly across different regions?
#Hypothesis: all regions have the same mean sales; Null: at least one region differs.
#Based off ANOVA , P>0.05 (0.0908>0.05) which means there is no significant difference between average
#sales across regions.

#Q2
model_category = smf.ols('total_revenue ~ C(product_category)',data=amazon_2023_first6).fit()
anova_category = sm.stats.anova_lm(model_category, typ=2)
print(anova_category)
```